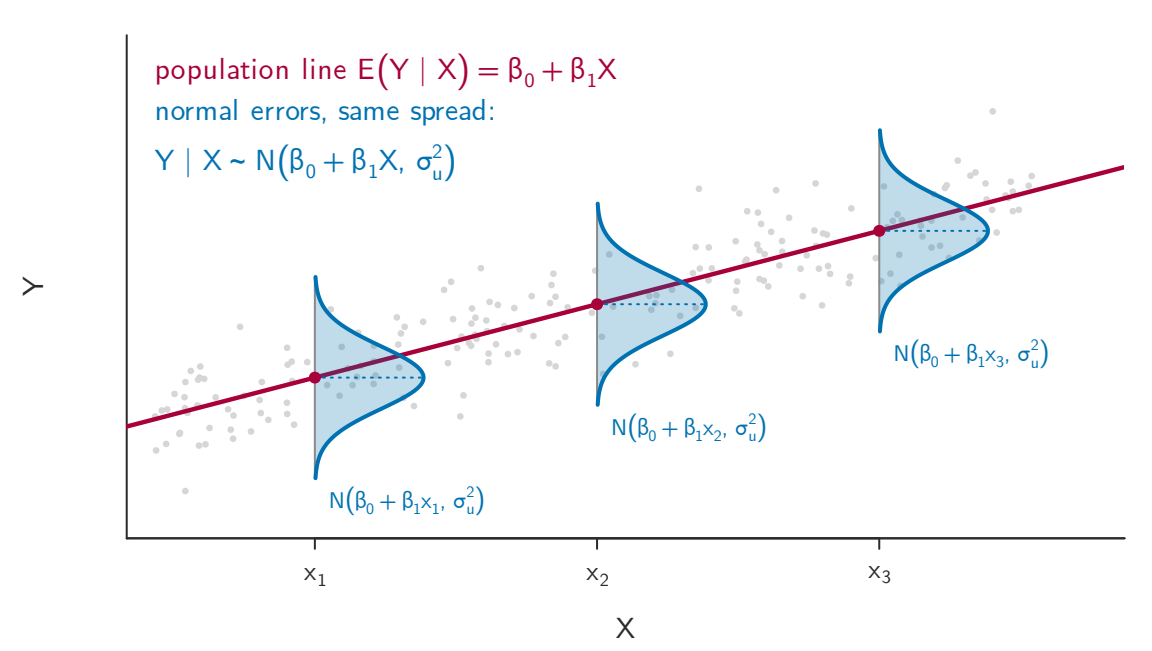
population line $E(Y \mid X) = \beta_0 + \beta_1 X$
normal errors, same spread:
$Y \mid X \sim N(\beta_0 + \beta_1 X, \sigma_u^2)$
$N(\beta_0 + \beta_1 x_1, \sigma_u^2)$
$N(\beta_0 + \beta_1 x_2, \sigma_u^2)$
$N(\beta_0 + \beta_1 x_3, \sigma_u^2)$
Y
X
x_1
x_2
x_3